MATH 110

Dani Nguyen

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1.1 Lecture 1 - Introduction to PDEs

What is a ODE? Recall that an ordinary differential equation (ODE) are equations for functions of a single variable, $u(t),\ t \in \mathbb{R}$

$$\frac{du}{dt} = u(1-u) \text{ is an ODE}$$

Here we call t the independent variable and u the state/dependent variable. In general, an ODE has the form $F(t) = (t, u, \frac{du}{dt}, \frac{d^2u}{dt^2}, \dots)$

Note that the state variable can have many components.

$$\mathbf{u}(t) = \begin{bmatrix} u_1(t) \\ u_2(t) \end{bmatrix} \in \mathbb{R}^2$$

then

$$\frac{du}{dt} = \mathbf{u} \text{ is an ODE}$$

What is a PDE? A PDE is an equation for a function with more than 1 independent variable, e.g. $u(x, y)$.

$$\text{Notation: } u_x = \frac{\partial u}{\partial x} = \partial_x u$$
$$u_y = \frac{\partial u}{\partial y} = \partial_y u$$
$$\text{etc.}$$

<u>Examples of PDEs</u>

- Heat/diffusion equation for $u(x, t)$

$$u_t = k u_{xx} \quad k \text{ constant}$$

- Wave equation for $u(x, t)$

$$u_{tt} = c^2 u_{xx} \quad c \text{ constant}$$

- Laplace's equation for $u(x, y)$

$$u_{xx} + u_{yy} = 0$$

$\boxed{\text{ex}}$ Consider the heat flow in a bar of length l

- $u(x, t)$ is the temperature of the bar at position x, time t
- Suppose the ends are held at temperature 0, and the initial temperature is $\phi(x)$
- The temperature evolves according to the heat equation $u_t = k u_{xx}$ where $t \geq 0, 0 \leq x \leq l$, k constant.

We also have:

- Boundary conditions (BCs):

$$u(0, t) = u(l, t) = 0 \quad \forall t$$

- Initial conditions (ICs):

$$u(x, 0) = \phi(x)$$

Remarks

- The order of the PDE is the order of the highest derivative that occurs.
- The heat equation above is 2^{nd} order
- A general 2^{nd} order PDE for $u(x,t)$ is:

$$G(x, t, u, u_x, u_t, u_{xx}, u_{tt}, u_{xt}) = 0$$

- A PDE in the above form is linear if G is a linear function of u and all of its derivatives. It is nonlinear otherwise.
- A linear PDE is homogeneous if every term contains u or some derivative of u. It is nonhomogeneous/inhomogeneous otherwise.

$\boxed{\text{ex}}$ Consider the linearity and homogeneity of the following equations.

1. $u_t + e^{x^2t}u_{xt} = 0$
   is a linear and homogeneous PDE.
2. $u_t + 3xu_{xx} = \frac{t}{x}$
   is a linear and inhomogeneous PDE because of the $\frac{t}{x}$ term.
3. $u_t + uu_x = 0$
   is a nonlinear PDE because of the uu_x term.
4. $u_{tt} + u_x + sin(u) = 0$
   is nonlinear because of the $sin(u)$ term.

Operating notation & linearity
We can write the heat equation $u_t - ku_{xx} = 0$ as $Lu = 0$ where

$$L = (\partial_t - k\partial_x^2)$$

We call L the differential operator. L acts on (smooth) functions $u(x,t)$ to produce a new function.

$$\begin{aligned} L(u) = Lu &= (\delta_t - k\partial_x^2)u \\ &= u_t - ku_{xx} \end{aligned}$$

L is a linear operator iff

$$\begin{aligned} L(u+v) &= Lu + Lv \\ &\quad \forall \text{ functions } u, v \end{aligned}$$

and

$$\begin{aligned} L(cu) &= cLu \\ &\quad \forall \text{ constants } c \in \mathbb{R} \\ &\quad \forall \text{ functions } u \end{aligned}$$

$\boxed{\text{ex}}$ Check if the heat equation has a linear differential operator.

1.

$$\begin{aligned} L(u+v) &= (\partial_t + k\partial_x^2)(u+v) \\ &= u_t + ku_{xx} + v_t + kv_{xx} \\ &= Lu + Lv \quad \checkmark \end{aligned}$$

2.

$$\begin{aligned} L(cu) &= cu_t + cku_{xx} \\ &= cLu \text{ for constant } c \quad \checkmark \end{aligned}$$

Remarks

- If L is a linear operator then $Lu = f$ is a linear PDE

$$\underbrace{u_t + ku_{xx}}_{Lu} = \underbrace{e^{x^2 t}}_{f(x,t)} \text{ is linear}$$

- If L is a linear operator then the PDE $Lu = f$ is
  - homogeneous if $f = 0$
  - inhomogeneous if $f \neq 0$
- A linear, homogeneous PDE $Lu = 0$ has the superposition principle

- If u_1, u_2 are solutions to the PDE then $au_1 + bu_2$ is a solution for constants a, b.

$$Lu_1 = Lu_2 = 0$$
$$\Rightarrow L(au_1 + bu_2) = 0$$

- The distinction between linear & nonlinear PDEs is very important.

Solutions to PDEs

- A solution to a PDE is a function u in which satisfies the PDE and any BCs/ICs upon direct substitution.
- The general solution to an ODE involves arbitrary constants. The general solution to a PDE involves arbitrary functions.
- Typically, we are interested in specific solutions.

$\boxed{\text{ex}}$ Consider the heat equation $u_t - u_{xx} = 0$.

$$u_1(x,t) = x^2 + 2t \text{ is a soln}$$
$$\rightarrow \partial_t u_1 - \partial_x^2 u_1 = 2 - 2 = 0$$
$$u_2(x,t) = e^{-t} \sin x \text{ is a soln}$$
$$\rightarrow \partial_t u_2 - \partial_x^2 u_2 = -e^{-t} \sin x + e^{-t} \sin x = 0$$

$\boxed{\text{ex}}$ Consider the equation $u_x = tx$ (1st order, linear, inhomogeneous)

$$Lu = f \text{ with } L = \partial_x$$
$$f(x,t) = tx$$

We can solve this by direct integration on both sides wrt x.

$$u(x,t) = \frac{1}{2} tx^2 + A(t)$$

Where $A(t)$ is an arbitrary function of t

$\boxed{\text{ex}}$ Consider the equation $u_{tt} - 4u = 0$ (2nd order, linear, homogeneous)

$$Lu = f$$
$$\rightarrow L = (\partial_t^2 - 4)$$
$$f(x,t) = 0$$

There is no explicit x-dependence

- We can treat this as an ODE with x as a parameter.
- 'Constants' depend on x
- General solution:

$$u(x,t) = A(x)e^{-2t} + B(x)e^{2t}$$

1.2 Lecture 2 -